2025

Hackaday Supercon

BLING IT ON!

AGENDA

- Who are we?
- A look at the hardware
- Visualizations and Generative Art
- Working with constraints
- Helpers and concepts
- CircuitPython Framework & Hello World
- Workflow
- APIs
- A journey through the example
- Make it your own / C++ version for nerds



ESTEFANNIE
Computer Scientist & Content Creator

@estefanniegg

Estefannie is an engineer who blends technology with social awareness to create thought-provoking pieces and makes satirical videos about them on her social media.



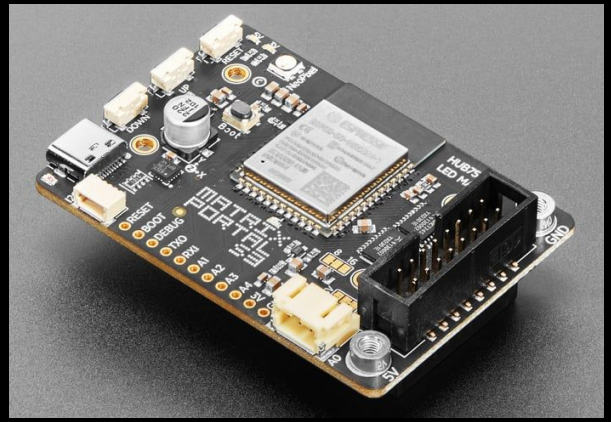
BOB (Instant Arcade)
Software & Electronics Nerd

leds.social/@instantarcade

Ex Disney Imagineer and Technology Director.
Bob spends a lot of his time tinkering with the fusion of software and hardware, primarily with very blinky results.

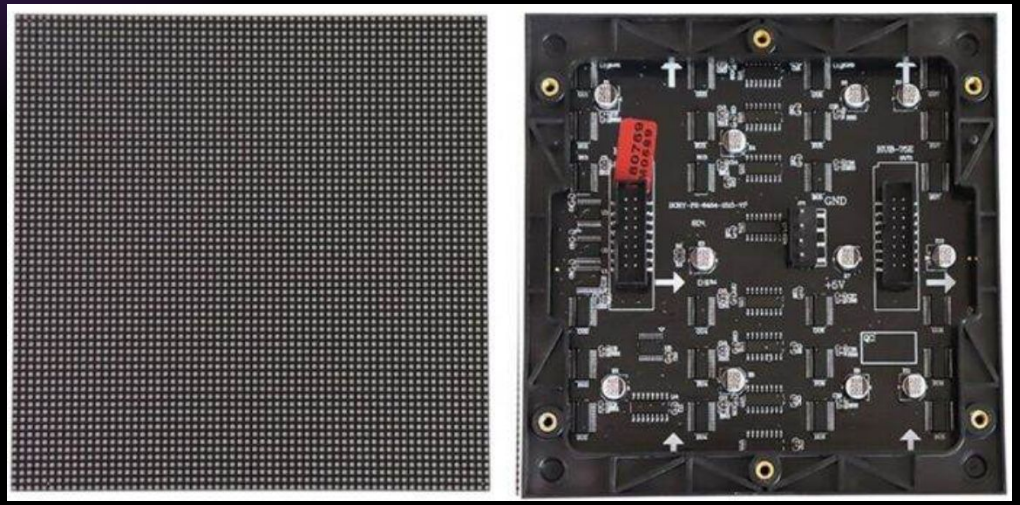
Hardware

The brains of the operation (aside from your own) is the Adafruit MatrixPortal S3 which is an ESP32-S3 based board with onboard level shifting for driving a HUB75 style matrix, LIS3DH Accelerometer, JST connector, GPIO Breakout, and built-in NeoPixel



Hardware

The other part of this dynamic duo is a 2mm dot pitch 64x64 RGB LED HUB75E type matrix panel which are widely used as segments for large displays



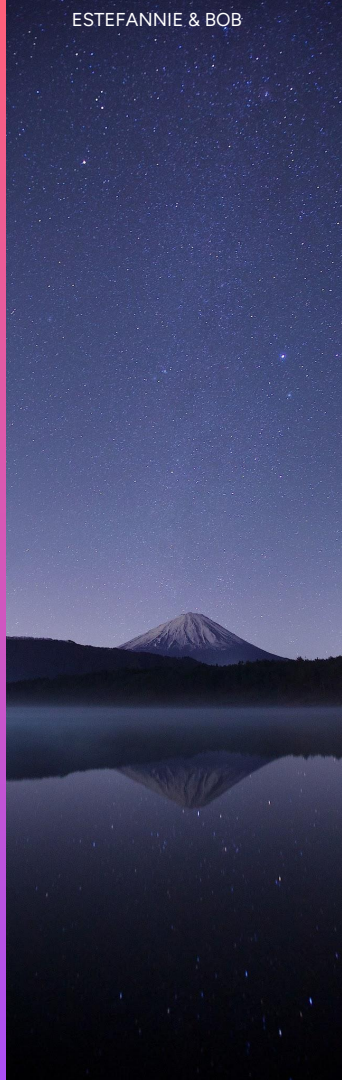
We’re also supplying power connectors and a ribbon cable to allow you to relocate the driver board if you don’t want it poking out of the case.

Visualizations & Generative Art

Visualizations are simply graphical representations of Data, Algorithms, or Both!

Generative Art is type of digital art that uses algorithms or other mathematical processes to create unique, dynamic pieces. These artworks often involve random elements and can produce an almost infinite number of variations, making each piece one-of-a-kind

For inspiration, check out:
Patt Vira - www.PattVira.com
Daniel Shiffman - www.TheCodingTrain.com



Working with constraints

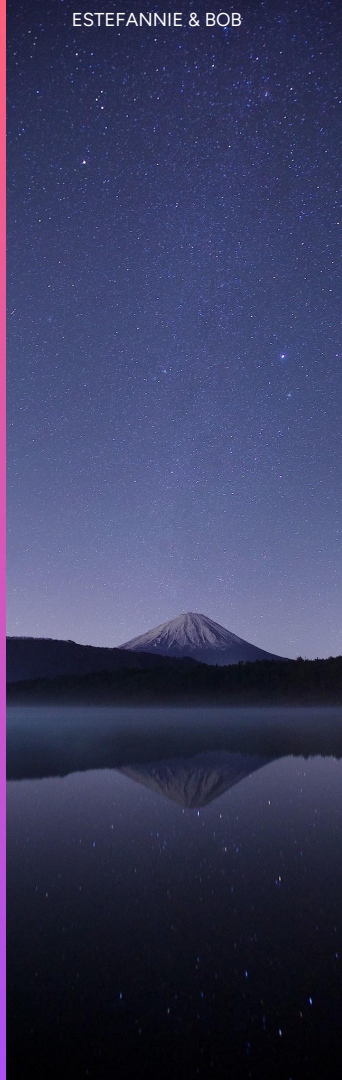
We have a few major constraints:

- Limited processing power
- Limited memory
- Small display areas (64x64 pixels)
- Every pixel set per update costs performance

With this in mind, we should be careful to choose techniques and algorithms that are less affected by these constraints

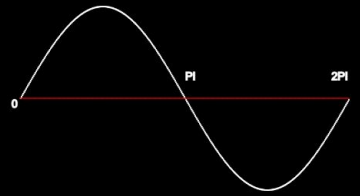
CircuitPython is an interpreted language making computations even slower.

Taking advantage of built in drawing commands will help us mitigate some of these concerns



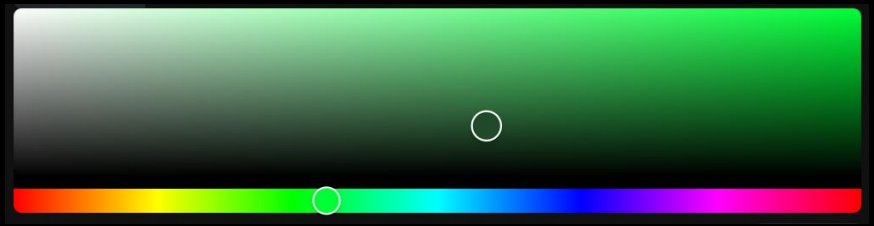
Helpers & Concepts

Sine waves are a fantastic way to get dynamic movement in a visualization for not much effort



Scaling everything to a 0-1 range makes manipulation much easier and more intuitive. We can easily scale back up to screen coordinates before displaying

Working in the HSV colorspace makes understanding and manipulating colors a much easier task. There's a helper to convert between the HSV and RGB values that are native to the display

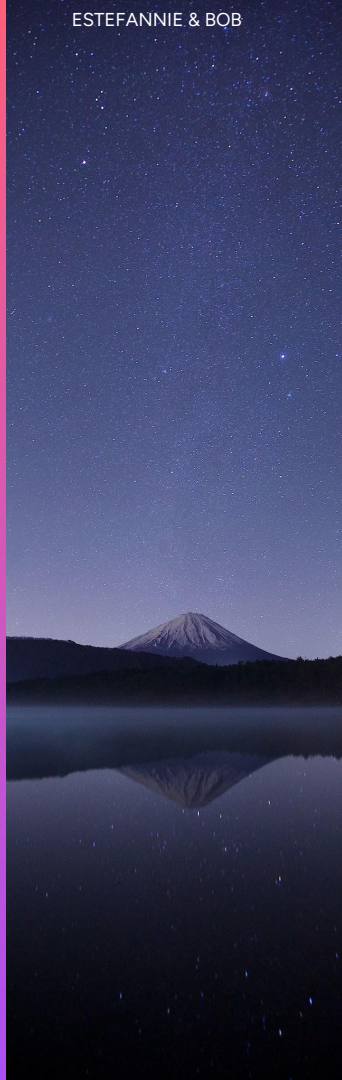


CircuitPython Framework & Hello World

The basic framework for this workshop puts all the initialization in code.py and uses a Visualization class to do the custom rendering.

Visualizations will be in their own files imported into the main code.py

Visualizations have a reset() method, plus an update() method called from the mainloop.



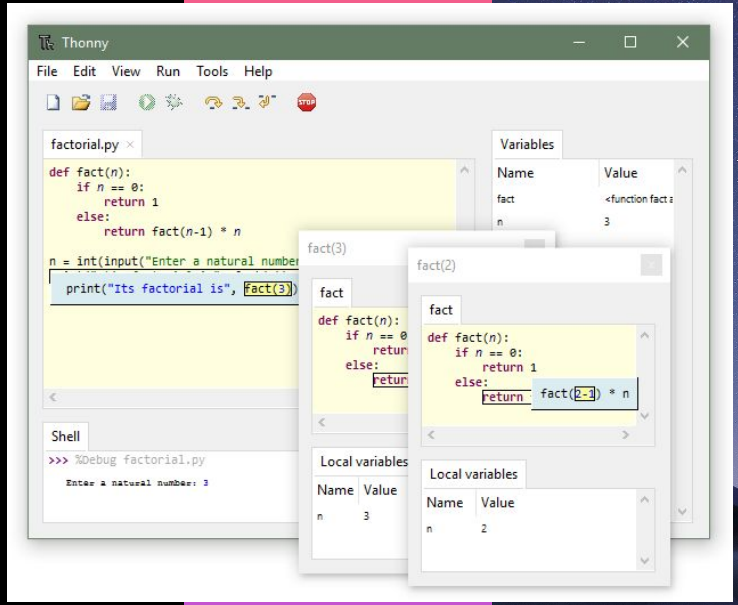
Workflow

We will use the Thonny editor
https://thonny.org

Best practice is to work locally and copy the updated files to the board before running them.

Ctrl-C to stop the currently running script.
Ctrl-S to save the file.
Use the menu to copy the script to the Micro
Select the code.py tab and hit F5 to run it

Keep an eye on the shell for errors or information you print()



Let's talk APIs!

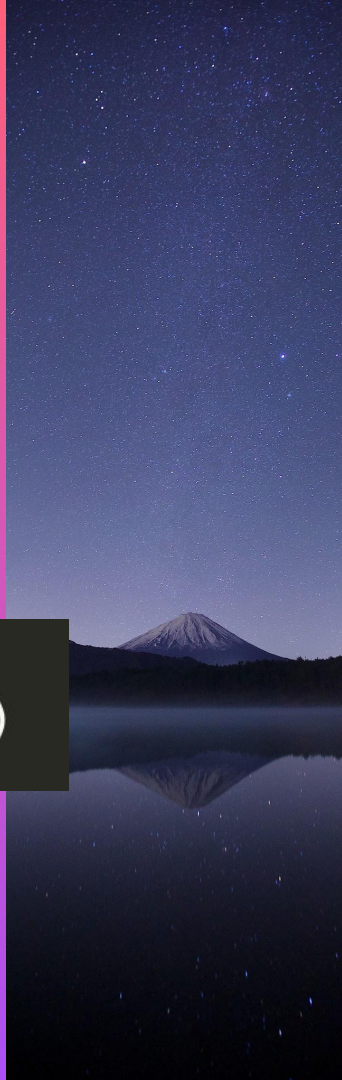
You can bring your visualizations to life with data from the internet

API (Application Programming Interface) is a way for one program to interact with another.

Most APIs return data in JSON which is easy to parse in Python
To get started, all you need is the requests library:

```
import requests
data = requests.get("https://api.example.com").json()
```

APIs let you pull live data like:
🌦️ weather 🪐 space 🎨 art 🧠 random facts



APIs visualization ideas

Earth vs Mars Weather

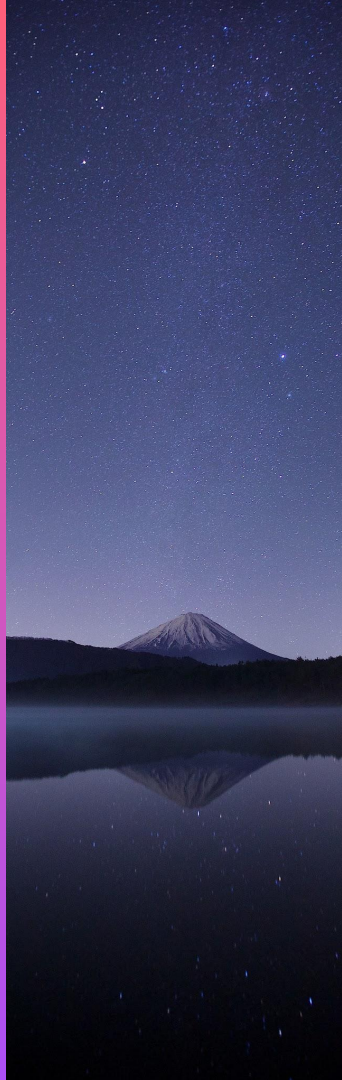
Use NASA's Mars Weather API to get daily atmospheric data from Elysium Planitia and compare it to real-time weather data from Los Angeles. Display the planets side by side 🌤️🪐

Clock

Create a custom clock that displays the current time, use the rest of the panel to make animated backgrounds, and match your aesthetic.

Handheld Planetarium

Use a Star API to get fetch real constellation positions, visualize them, and use the accelerometer to look around the sky.



Follow us and join the Bling it On! Discord



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You can also find us at electronics club at Social Medium Makerspace
www.socialmedium.space